

# Advanced Data Analysis using Python

## UNIT I: Introduction to Data Analysis & Python Libraries

1. What is data analysis? Explain its steps.
2. Write a note on the importance of Python in data analysis.
3. Explain the features of NumPy and its key operations.
4. Describe how to create and manipulate arrays using NumPy.
5. What is Pandas? How are Series and DataFrames created and used?

## UNIT II: Data Wrangling with Pandas

1. Explain data cleaning techniques using Pandas.
2. How do you handle missing data in a DataFrame?
3. Describe techniques for data transformation (filtering, mapping, replacing).
4. How can data be merged, joined, and concatenated in Pandas?
5. Explain grouping and aggregation in Pandas with examples.

## UNIT III: Data Visualization

1. What is the role of data visualization in analysis?
2. Explain different types of plots using Matplotlib (line, bar, histogram, scatter).
3. Write a short note on Seaborn and its advantages over Matplotlib.
4. How do you create subplots in Matplotlib?
5. Explain how to customize plots (labels, titles, legends).

## UNIT IV: Statistical Analysis using Python

1. Describe descriptive statistics (mean, median, mode, std, etc.) in Python.
2. Explain how to perform correlation and covariance analysis.
3. What is hypothesis testing? Explain using Python (t-test, chi-square).
4. Write Python code to perform linear regression.
5. Explain ANOVA and its application in Python.

## UNIT V: Advanced Analysis & Case Studies

1. Describe time series analysis with Pandas.
2. What is Principal Component Analysis (PCA)? Explain with an example.
3. Explain clustering techniques (like K-means) using Python.
4. Discuss a real-life case study using Python for data analysis.
5. What are the best practices in advanced data analysis?